

QuadRotor Helicopter Modeling

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2.1 Introduction

This chapter looks into the modeling of the *QuadRotor* autonomous aerial vehicle. The equations of motion are developed based on physical laws that describe the location of the mass center of the vehicle, using three position coordinates, and three Euler angles to specify the orientation of the body. Two approaches are used to obtain the dynamic equations that describe explicitly the relationship between force and motion. First the Euler-Lagrange formulation is described. Afterwards, an alternative description of the autonomous aerial vehicle is obtained using the Newton-Euler formulation.

The autonomous aerial vehicle used in this thesis is a miniature helicopter in a coplanar configuration of four rotors (*QuadRotor*), as illustrated in Fig. 2.1. In this figure, the forces generated by each propeller to produce motion can be also observed. The movement of the UAV results from changes on the lift force caused

by adjusting the velocities of each rotor. Longitudinal motions (i.e. pitch motions) are achieved varying the front and rear rotors velocity, which change the forces f_1 and f_3 (see Fig. 2.1), while lateral displacements (i.e. roll motions) are performed through the speed variation of the right and left propellers, which vary the forces f_2 and f_4 . Yaw movements are obtained from the difference in the counter-torque between each pair of propellers, (f_1, f_3) and (f_2, f_4) , i.e., accelerating the two clockwise turning rotors while decelerating the counter-clockwise turning rotors, and vice-versa. This movement is possible because the rotors 1 and 3 rotate in the opposite direction to the rotors 2 and 4. Finally, the total thrust T , which displaces the helicopter in the perpendicular plane with respect to the propellers, is obtained by the sum of the four forces generated by propellers.

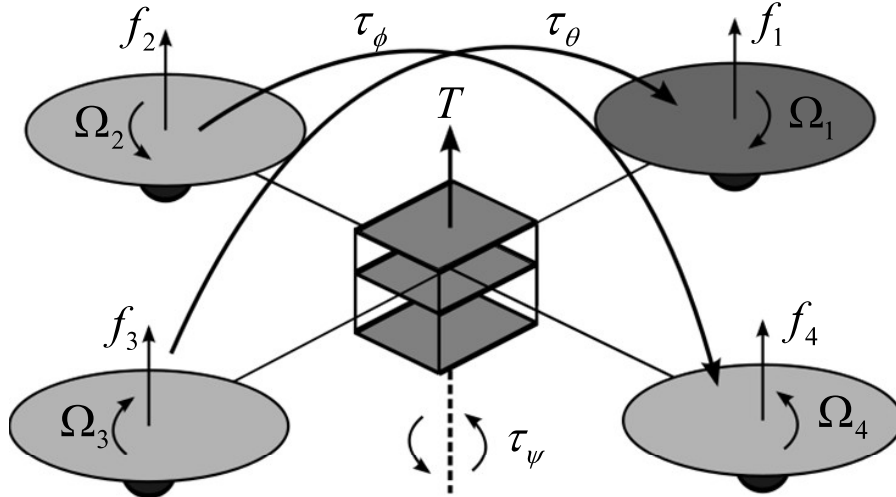


Figure 2.1: Operating diagram of the *QuadRotor* helicopter.

Therefore, the *applied* thrust, T , is given by (Castillo et al., 2005b):

$$T = \left(\sum_{i=1}^4 f_i \right) = \left(\sum_{i=1}^4 b \Omega_i^2 \right), \quad (2.1)$$

where f_i is the force generated by each rotor, Ω_i is the angular velocity of the i th rotor around its axis and b is the thrust coefficient of the rotors. Moreover, the *applied* torque vector on the three axes is given by:

$$\boldsymbol{\tau}_a = \begin{bmatrix} \tau_{\phi_a} \\ \tau_{\theta_a} \\ \tau_{\psi_a} \end{bmatrix} = \begin{bmatrix} (f_2 - f_4)l \\ (f_3 - f_1)l \\ \sum_{i=1}^4 \tau_{M_i} \end{bmatrix}, \quad (2.2)$$

where l is the distance between the rotors and the center of rotation, and τ_{M_i} is the torsion effort generated by each electrical motor, considering the dynamic of each disc of the motor as an uncoupled system in the generalized variable Ω_i . The torsion effort of the motor is opposite to its aerodynamic force $\tau_{drag} = k_\tau \Omega_i^2$, where $k_\tau > 0$ is a constant. Thus, by using the Newton's second law, the torque generated by each rotor is given by (Castillo et al., 2005b):

$$J_R \dot{\Omega}_i = -\tau_{drag} + \tau_{M_i}, \quad (2.3)$$

and keeping the total thrust constant, that is, considering $\dot{\Omega}_i = 0$, it follows that:

$$\tau_{M_i} = \tau_{drag} = k_\tau \Omega_i^2. \quad (2.4)$$

Then, the equation of the *applied* torques (2.2) can be rewritten as follows:

$$\begin{bmatrix} \tau_{\phi_a} \\ \tau_{\theta_a} \\ \tau_{\psi_a} \end{bmatrix} = \begin{bmatrix} lb(\Omega_2^2 - \Omega_4^2) \\ lb(\Omega_3^2 - \Omega_1^2) \\ k_\tau(\Omega_1^2 + \Omega_3^2 - \Omega_2^2 - \Omega_4^2) \end{bmatrix}. \quad (2.5)$$

Additionally, this kind of system is a flight vehicle of lightweight structure and, therefore, gyroscopic effects resulting from the rotation of the rigid body and the four propellers should be included in the dynamic model (Bouabdallah et al., 2004b). Assuming each rotor as a rigid disc rotating around its vertical axis with an angular rate Ω_i , and that the rotation axis of the rotor rotates with the angular velocity of the reference frame, the gyroscopic moments due to the change in the orientation of the propeller plane are given by:

$$\tau_G = - \sum_{i=1}^4 J_R (\boldsymbol{\omega} \times \mathbf{e}_3) \cdot \Omega_i, \quad (2.6)$$

where J_R is the inertia moment of the motor around its rotation axis, $\boldsymbol{\omega}$ is the angular velocity of the helicopter, expressed in the body-fixed frame and $\mathbf{e}_3 = [0 \ 0 \ 1]'$.¹ However, in this thesis the dynamic model of the system is obtained under the assumption that the vehicle is a rigid body in the space, subject to one main force (thrust) and three torques. This simplification implies that gyroscopic effects caused by the propellers will be considered as disturbances for the rotational control law.

Besides, this helicopter is an underactuated mechanical system with six de-

¹The notation prime ' denotes transpose.

degrees of freedom and only four control inputs, which increases the complexity and proposes an enormous challenge to the control area. Due to this difficulty, some assumptions are made to compute the model for control purposes. The ground effect is neglected and the helicopter airframe is assumed to be symmetric, which results in a moment of inertia tensor of the center of the body-fixed frame with just diagonal inertia terms.

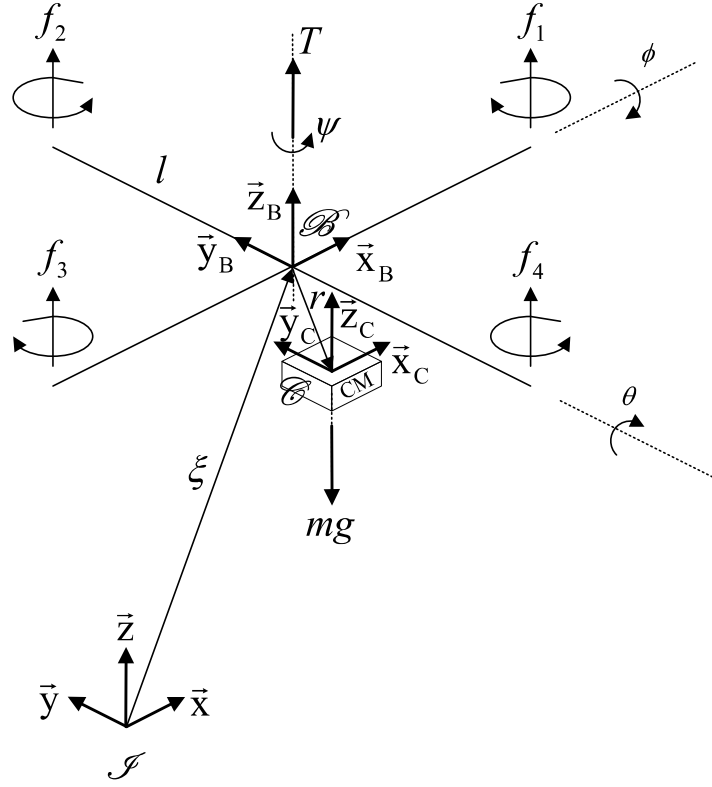
Assuming these hypotheses, two approximations of the helicopter model are presented. First, a more realistic model is obtained by using the Steiner's parallel-axis theorem. This model considers that the axes passing through the center of mass are parallel to the axes of rotation of the body-fixed frame, and its origin is displaced by a position $\mathbf{r}$ from the origin of the same frame. This assumption results in a strongly-coupled dynamic model. To overcome that, a second model is proposed, where the center of mass and the body-fixed frame origin are assumed congruent, which leads to a decentralized dynamic model.

In what follows, the helicopter attitude is defined, which makes possible to obtain the helicopter kinematics of the translational and rotational motions. A brief explanation of the well-known Euler-Lagrange formulation is given, and based on that, the helicopter equations of motion are obtained. To complete this chapter, the equations of motion of the *QuadRotor* are presented by means of the Newton-Euler procedure.

2.2 Helicopter Attitude

Since the equations of motion depend on the attitude of the mechanical system, this section presents how to estimate the position and the orientation of the rigid body according to an inertial reference frame.

The helicopter, as a rigid body, is characterized by a frame linked to it, and with the origin at its center of rotation. Furthermore, assuming the center of mass is displaced by a position $\mathbf{r} = [r_x \ r_y \ r_z]'$ from the origin of the rotation body-fixed frame and expressed in this frame, three perpendicular axes passing through the center of mass and parallel to the body-fixed frame are considered ($\mathcal{C} = \{\vec{x}_C, \vec{y}_C, \vec{z}_C\}$). Let $\mathcal{B} = \{\vec{x}_B, \vec{y}_B, \vec{z}_B\}$ be the rotation body-fixed frame, where the $\vec{x}_B$ axis is the helicopter normal flight direction, $\vec{y}_B$ is orthogonal to $\vec{x}_B$ and negative to starboard in the horizontal plane, whereas $\vec{z}_B$ is oriented in ascendant sense and orthogonal to the plane $\vec{x}_B O \vec{y}_B$. The inertial frame $\mathcal{I} = \{\vec{x}, \vec{y}, \vec{z}\}$ is considered fixed with respect to the earth (see Fig. 2.2).

Figure 2.2: *QuadRotor* helicopter scheme.

To compute the coordinates of a point $\mathbf{p}$ rigidly attached to the frame $\mathcal{C}$ relative to the frame $\mathcal{I}$, with local coordinates ${}^{\mathcal{C}}\mathbf{p}$, the following equations must be composed:

$${}^{\mathcal{B}}\mathbf{p} = {}^{\mathcal{B}}\mathbf{R}_{\mathcal{C}} {}^{\mathcal{C}}\mathbf{p} + {}^{\mathcal{B}}\mathbf{r}, \quad (2.7)$$

$${}^{\mathcal{I}}\mathbf{p} = {}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}} {}^{\mathcal{B}}\mathbf{p} + {}^{\mathcal{I}}\boldsymbol{\xi}, \quad (2.8)$$

resulting in:

$${}^{\mathcal{I}}\mathbf{p} = {}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}} {}^{\mathcal{B}}\mathbf{R}_{\mathcal{C}} {}^{\mathcal{C}}\mathbf{p} + {}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}} {}^{\mathcal{B}}\mathbf{r} + {}^{\mathcal{I}}\boldsymbol{\xi}, \quad (2.9)$$

where the rotation matrices ${}^{\mathcal{B}}\mathbf{R}_{\mathcal{C}}$ and ${}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}}$ specify the orientations of frame $\mathcal{C}$ with respect to $\mathcal{B}$, and of this last one with respect to the frame $\mathcal{I}$, respectively. The vector ${}^{\mathcal{I}}\boldsymbol{\xi} = \boldsymbol{\xi} = [x \ y \ z]'$ represents the position of the origin of the rotation frame of the helicopter, expressed in the inertial frame $\mathcal{I}$. The vector ${}^{\mathcal{B}}\mathbf{r} = \mathbf{r}$, as commented above, describes the position of the center of mass, which is located at the origin of the frame $\mathcal{C}$, with respect to the origin of the frame $\mathcal{B}$.

As frames $\mathcal{C}$ and $\mathcal{B}$ are parallel, the rotation matrix ${}^{\mathcal{B}}\mathbf{R}_{\mathcal{C}}$ is equal to the identity matrix, which results in ${}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}} = {}^{\mathcal{I}}\mathbf{R}_{\mathcal{C}}$. For the sake of notation simplicity, both rotation matrices ${}^{\mathcal{I}}\mathbf{R}_{\mathcal{B}}$ and ${}^{\mathcal{I}}\mathbf{R}_{\mathcal{C}}$ are defined as $\mathbf{R}_{\mathcal{I}}$. Therefore, equation

(2.9) can be written in a compact form:

$${}^{\mathcal{I}}\mathbf{p} = \mathbf{R}_{\mathcal{I}}^{\mathcal{C}} {}^{\mathcal{C}}\mathbf{p} + {}^{\mathcal{I}}\mathbf{d}_{\mathcal{C}}, \quad (2.10)$$

which describes the rigid motion between ${}^{\mathcal{I}}\mathbf{p}$ and ${}^{\mathcal{C}}\mathbf{p}$ (Spong et al., 2006). The vector from the origin of the frame $\mathcal{I}$ to the origin of the frame $\mathcal{C}$ is given by:

$${}^{\mathcal{I}}\mathbf{d}_{\mathcal{C}} = \mathbf{R}_{\mathcal{I}} \mathbf{r} + \boldsymbol{\xi}, \quad (2.11)$$

and the vehicle orientation is given by the rotation matrix $\mathbf{R}_{\mathcal{I}} : \mathcal{B}, \mathcal{C} \rightarrow \mathcal{I}$, where $\mathbf{R}_{\mathcal{I}} \in SO(3)$ is an orthonormal rotation matrix.

The equation of the rigid motion (2.10) can be represented in a matrix form through the homogeneous transformation matrix as follows:

$${}^{\mathcal{I}}\bar{\mathbf{p}} = \underbrace{\begin{bmatrix} \mathbf{R}_{\mathcal{I}} & {}^{\mathcal{I}}\mathbf{d}_{\mathcal{C}} \\ \mathbf{0}_{1 \times n} & 1 \end{bmatrix}}_{\mathbf{H}_{\mathcal{I}}} \cdot {}^{\mathcal{C}}\bar{\mathbf{p}}, \quad (2.12)$$

where ${}^{\mathcal{C}}\bar{\mathbf{p}} = [x_C \ y_C \ z_C \ 1]'$.

The rotational transformation of a UAV or, in broad terms, of a rigid body, can be parameterized through several methods such as: the Euler-angle representation, the quaternions representation, and the axis/angle representation (Spong et al., 2006). From twelve independent definitions of the Euler angles is possible to represent the relative orientation between two coordinate systems. The most common are the *x-convention* (rotation around z, x, z), the *y-convention* (rotation around z, y, z) and the *xyz-convention* (rotation around x, y, z). The last one, using intrinsic rotations, is usually found on aerospace applications, and it is named roll, pitch and yaw representation, also known as Tait-Bryan convention, Cardano convention, or nautical convention.

The rotation matrix can be obtained through three successive rotations around the axes of the body-fixed frame. In this thesis, the roll, pitch and yaw angles are used to describe the helicopter rotation in the three-dimensional Euclidean space with respect to the body-fixed frame. Thereby, the configuration of a rigid body rotation in the space is performed as follows (Murray et al., 1994):

1. *Rotation around $\vec{z}_B$ by ψ* : the first movement, assuming the inertial and the body-fixed frames coincide, is given by a rotation around the $\vec{z}_B$ axis from

the $\vec{z}$ axis by the *yaw* angle, ψ .

$$\begin{bmatrix} x'_B \\ y'_B \\ z'_B \end{bmatrix} = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (2.13)$$

2. Rotation around $\vec{y}_B$ by θ : the second movement is performed by a *pitch* motion around the $\vec{y}_B$ from the new $\vec{y}_B$ axis.

$$\begin{bmatrix} x''_B \\ y''_B \\ z''_B \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} x'_B \\ y'_B \\ z'_B \end{bmatrix} \quad (2.14)$$

3. *Rotation around $\vec{x}_B$ by ϕ* : the third movement is a rotation that corresponds to the *roll* angle, ϕ , it is carried out around the $\vec{x}_B$ axis from the new $\vec{x}_B$ to move the helicopter to the final position.

$$\begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} x''_B \\ y''_B \\ z''_B \end{bmatrix} \quad (2.15)$$

These angles are bounded as follows: roll angle, ϕ , by $(-\pi < \phi < \pi)$; pitch angle, θ , by $(-\pi/2 < \theta < \pi/2)$; and yaw angle, ψ , by $(-\pi < \psi < \pi)$.

From these three movements, the following rotation matrices that represent the orientation of the rigid body rotating around of each axis are defined:

$$\mathbf{R}(x_B, \phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}, \quad \mathbf{R}(y_B, \theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix},$$

$$\mathbf{R}(z_B, \psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Note that the rotations matrices $\mathbf{R}(x_B, \phi)$, $\mathbf{R}(y_B, \theta)$ and $\mathbf{R}(z_B, \psi)$ describe rotations starting from the inertial frame to the body-fixed one. Thus, the rotations from the body-fixed frame to the inertial one are given by: $\mathbf{R}(x, \phi) = \mathbf{R}(x_B, \phi)'$, $\mathbf{R}(y, \theta) = \mathbf{R}(y_B, \theta)'$ and $\mathbf{R}(z, \psi) = \mathbf{R}(z_B, \psi)'$

The following rotation matrix from $\mathcal{I}$ to $\mathcal{B}$ (it is the same one from $\mathcal{I}$ to $\mathcal{C}$),

named as Direction-Cosine Matrix, is obtained by:

$$\begin{aligned} \mathbf{R}_{\mathcal{B}} &= {}^{\mathcal{B}}\mathbf{R}_{\mathcal{I}} = \mathbf{R}(x_B, \phi) \cdot \mathbf{R}(y_B, \theta) \cdot \mathbf{R}(z_B, \psi) \\ \mathbf{R}_{\mathcal{B}} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \mathbf{R}_{\mathcal{B}} &= \begin{bmatrix} C\psi C\theta & S\psi C\theta & -S\theta \\ C\psi S\theta S\phi - S\psi C\phi & S\psi S\theta S\phi + C\psi C\phi & C\theta S\phi \\ C\psi S\theta C\phi + S\psi S\phi & S\psi S\theta C\phi - C\psi S\phi & C\theta C\phi \end{bmatrix}, \end{aligned}$$

where $C \cdot = \cos(\cdot)$ and $S \cdot = \sin(\cdot)$.

The rotation matrix that describes the orientation of the body-fixed frame with respect to the inertial frame $\mathcal{I}$ is the transpose matrix of $\mathbf{R}_{\mathcal{B}}$, due to its orthonormality property, and is given by:

$$\mathbf{R}_{\mathcal{I}} = \begin{bmatrix} C\psi C\theta & C\psi S\theta S\phi - S\psi C\phi & C\psi S\theta C\phi + S\psi S\phi \\ S\psi C\theta & S\psi S\theta S\phi + C\psi C\phi & S\psi S\theta C\phi - C\psi S\phi \\ -S\theta & C\theta S\phi & C\theta C\phi \end{bmatrix}. \quad (2.16)$$

2.3 Helicopter Kinematics

The kinematic equations of the rotational and translational movements are obtained by means of the rotation matrix (2.16) and the equation of rigid motion (2.10).

Since the frame $\mathcal{C}$ is attached to $\mathcal{B}$, the angular velocity of the frame $\mathcal{C}$ is the same as the angular velocity of the rotation body-fixed frame. Thus, the rotational kinematics can be obtained from the relationship between the rotation matrix (2.16) and its time derivative with a skew-symmetric matrix, which simplifies many of the computations involved (Craig, 1989; Spong et al., 2006).

Let $\mathbf{R} \in \mathfrak{R}^{n \times n}$ be an orthonormal matrix, where:

$$\mathbf{R}'\mathbf{R} = \mathbf{1}_{n \times n}. \quad (2.17)$$

Its time derivative is given by:

$$\dot{\mathbf{R}}'\mathbf{R} + \mathbf{R}'\dot{\mathbf{R}} = \mathbf{0}_{n \times n}. \quad (2.18)$$

By defining:

$$\mathbf{S} = \mathbf{R}'\mathbf{R}, \quad (2.19)$$

from (2.18) it is obtained that:

$$\mathbf{S}' + \mathbf{S} = \mathbf{0}_{n \times n}, \quad (2.20)$$

where $\mathbf{S}$ is a skew-symmetric matrix. Multiplying both sides of equation (2.19) on the left by $\mathbf{R}$ and using the fact that $\mathbf{R}\mathbf{R}' = \mathbf{1}_{n \times n}$ produces:

$$\dot{\mathbf{R}} = \mathbf{R}\mathbf{S}. \quad (2.21)$$

Therefore, the kinematic equation used to determine the helicopter rotational motion, assuming the rotation matrix (2.16), is given by:

$$\dot{\mathbf{R}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \cdot \mathbf{S}(\boldsymbol{\omega}), \quad (2.22)$$

where $\boldsymbol{\omega} = [p \ q \ r]'$ is the absolute angular velocity vector of the rigid body expressed in the rotation body-fixed frame (i.e., $\boldsymbol{\omega}$ is the angular velocity that corresponds to the derivative of ${}^{\mathcal{J}}\mathbf{R}_{\mathcal{B}}$, expressed in coordinates relative to the body-fixed frame $\mathcal{B}$), and $\mathbf{S}(\boldsymbol{\omega})$ ($\mathbf{S}(\boldsymbol{\omega})(\cdot) = \boldsymbol{\omega} \times \cdot$) is the following skew-symmetric matrix:

$$\mathbf{S}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}. \quad (2.23)$$

On the other hand, the angular velocity vector of the helicopter expressed in the inertial frame is given by:

$$\boldsymbol{\omega}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega}. \quad (2.24)$$

The relationship between the angular rates of the body-fixed frame and the time derivative of the Euler angles can be obtained from the equation (2.22) after some algebraic manipulations, or as follows:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + \mathbf{R}(x, \phi)^{-1} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + (\mathbf{R}(y, \theta)\mathbf{R}(x, \phi))^{-1} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix},$$

$$\boldsymbol{\omega} = \mathbf{W}_{\boldsymbol{\eta}} \dot{\boldsymbol{\eta}},$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \quad (2.25)$$

where $\boldsymbol{\eta} = [\phi \ \theta \ \psi]'$, and $\mathbf{W}_{\boldsymbol{\eta}}$ is the Euler matrix.

The helicopter rotational motion is given by the angular rate components in the three axis: the roll angular rate (p), the pitch angular rate (q), and the yaw angular rate (r), around the $\vec{x}_B$, $\vec{y}_B$ and $\vec{z}_B$ axes, respectively. These rotational velocities occur due to the torques applied on the system linked to the helicopter structure, which are generated by external forces. These external forces define the different moments in the three axes: rolling moment (τ_{ϕ_a}), pitching moment (τ_{θ_a}), and yawing moment (τ_{ψ_a}) around the $\vec{x}_B$, $\vec{y}_B$ and $\vec{z}_B$ axes, respectively (Esteban, 2005).

The time derivative of the Euler angles is obtained through the inverted Euler matrix in (2.25), and can be posed by:

$$\dot{\boldsymbol{\eta}} = \mathbf{W}_{\boldsymbol{\eta}}^{-1} \boldsymbol{\omega}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \quad (2.26)$$

The time derivative of the Euler angles ($\dot{\phi}, \dot{\theta}, \dot{\psi}$) is a discontinuous function. These time derivatives are different from the angular rates of the body-fixed frame (p, q, r), which can be physically measurable, for example, by means of gyroscopes. Generally, aerospace systems are equipped with Inertial Measurement Units (IMU's), that measure the angular rates and estimate the Euler angles (Bouabdallah et al., 2006).

The linear velocity of a point $\mathbf{p}$, that is rigidly attached to the frame $\mathcal{C}$ is obtained through the equation (2.10), where the velocity ${}^{\mathcal{I}}\dot{\mathbf{p}}$ is given by the product rule for differentiation as follows:

$${}^{\mathcal{I}}\dot{\mathbf{p}} = \dot{\mathbf{R}}_{\mathcal{I}}^{\mathcal{C}} \mathbf{p} + \mathbf{R}_{\mathcal{I}}^{\mathcal{C}} \dot{\mathbf{p}} + {}^{\mathcal{I}}\dot{\mathbf{d}}_{\mathcal{C}}, \quad (2.27)$$

where

$${}^{\mathcal{I}}\dot{\mathbf{d}}_{\mathcal{C}} = \dot{\mathbf{R}}_{\mathcal{I}} \mathbf{r} + \dot{\boldsymbol{\xi}}, \quad (2.28)$$

and as the point ${}^{\mathcal{C}}\mathbf{p}$ is assumed rigidly attached to the frame $\mathcal{C}$, its coordinates relative to the frame $\mathcal{C}$ do not change, giving ${}^{\mathcal{C}}\dot{\mathbf{p}} = 0$. Therefore, by substituting

the equations (2.22) and (2.28) into (2.27), the linear velocity is given by:

$$\begin{aligned}\mathcal{I}\dot{\mathbf{p}} &= \mathbf{R}_{\mathcal{I}}\mathbf{S}(\boldsymbol{\omega})^{\mathcal{C}}\mathbf{p} + \mathbf{R}_{\mathcal{I}}\mathbf{S}(\boldsymbol{\omega})\mathbf{r} + \dot{\boldsymbol{\xi}} \\ &= \mathbf{R}_{\mathcal{I}}\boldsymbol{\omega} \times^{\mathcal{C}}\mathbf{p} + \mathbf{R}_{\mathcal{I}}\boldsymbol{\omega} \times \mathbf{r} + \dot{\boldsymbol{\xi}},\end{aligned}\quad (2.29)$$

where $\mathbf{R}_{\mathcal{I}}\boldsymbol{\omega} \times^{\mathcal{C}}\mathbf{p}$ is the velocity of the point $\mathbf{p}$ with respect to the origin of the frame $\mathcal{C}$ expressed in the orientation of the frame $\mathcal{I}$, and the term $\boldsymbol{\omega} \times \mathbf{R}_{\mathcal{I}}\mathbf{r} + \dot{\boldsymbol{\xi}}$ is the rate at which the center of mass is moving. From now on, the linear velocity is defined by the simpler notation $\mathbf{v}_{\mathcal{I}} = \mathcal{I}\dot{\mathbf{p}}$.

The linear velocity of the point $\mathbf{p}$ expressed in the frame $\mathcal{C}$ is obtained by:

$$\mathbf{v}_{\mathcal{C}} = \mathbf{R}'_{\mathcal{I}}\mathbf{v}_{\mathcal{I}} = \mathbf{R}_{\mathcal{B}}\mathbf{v}_{\mathcal{I}}, \quad (2.30)$$

where $\mathbf{v}_{\mathcal{I}} = [u_0 \quad v_0 \quad w_0]'$ and $\mathbf{v}_{\mathcal{C}} = [u_C \quad v_C \quad w_C]'$.

2.4 Euler-Lagrange Mechanical Systems

In this section, the forced Euler-Lagrange equations of mechanical systems are presented.

Simple mechanical systems can be described by their Lagrangian, which is the difference between the (positive semidefinite) kinetic energy and the potential energy:

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = \mathcal{K}(\mathbf{q}, \dot{\mathbf{q}}) - \mathcal{U}(\mathbf{q}) = \frac{1}{2}\dot{\mathbf{q}}'\mathbf{M}(\mathbf{q})\dot{\mathbf{q}} - \mathcal{U}(\mathbf{q}), \quad (2.31)$$

where $\mathbf{q} \in \mathbb{Q}$ denotes the configuration vector of the system (i.e. the set of the so-called generalized coordinates) that belongs to an n -dimensional configuration manifold $\mathbb{Q}$, with n being the number of degrees of freedom of the system. $\mathcal{K}(\mathbf{q}, \dot{\mathbf{q}})$ is the system kinetic energy and $\mathcal{U}(\mathbf{q})$ is the potential energy, which here is related only to conservative forces such as the gravitational one.

$\mathbf{M}(\mathbf{q})$ is the inertia matrix which is a positive definite symmetric matrix. For a fixed value of the generalized coordinate $\mathbf{q}$, let $0 < \lambda_1(\mathbf{q}) \leq \dots \leq \lambda_n(\mathbf{q})$ denote the n eigenvalues of $\mathbf{M}(\mathbf{q})$. These eigenvalues are positive as a consequence of the positive definiteness of $\mathbf{M}(\mathbf{q})$ (Spong et al., 2006). It results that:

$$\lambda_1(\mathbf{q})\mathbf{1}_{n \times n} \leq \mathbf{M}(\mathbf{q}) \leq \lambda_n(\mathbf{q})\mathbf{1}_{n \times n}. \quad (2.32)$$

If all joints are of revolution, the inertia matrix contains only terms involving sine and cosine functions and, hence, is bounded above and below as a function of the

generalized coordinates. For a more conservative result, constants can be found to provide uniform bounds in the inertia matrix, whose magnitudes can be bounded by unity:

$$\lambda_{\min} \mathbf{1}_{n \times n} \leq \mathbf{M}(\mathbf{q}) \leq \lambda_{\max} \mathbf{1}_{n \times n}, \quad (2.33)$$

with $\lambda_{\min}$ and $\lambda_{\max}$ scalars. The boundedness property of the inertia matrix can also be defined as (Lewis et al., 2004):

$$0 < m_{\min} \leq \|\mathbf{M}(\mathbf{q})\| \leq m_{\max} \quad \forall \quad \mathbf{q} \in \mathbb{Q}, \quad (2.34)$$

where any induced matrix norm can be used to define the positive scalars $m_{\min}$ and $m_{\max}$.

Let $\mathbf{b}_i(\mathbf{q}) : \mathbb{Q} \rightarrow \mathbb{R}^n$, $i = \{1, \dots, m\}$ be an m vectors set of the external forces applied to the system. The Euler-Lagrange equation that describes this mechanical system is given by:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})}{\partial \mathbf{q}} = \mathcal{F}(\mathbf{q}), \quad (2.35)$$

where $\mathcal{F}(\mathbf{q}) \in \mathbb{R}^n$ is the generalized force/torque vector. By substituting (2.31) into (2.35), the following equation is obtained:

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{\mathbf{q}}} \left(\frac{1}{2} \dot{\mathbf{q}}' \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \right) \right) - \left(\frac{\partial}{\partial \mathbf{q}} \left(\frac{1}{2} \dot{\mathbf{q}}' \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \right) \right) + \frac{\partial \mathcal{U}(\mathbf{q})}{\partial \mathbf{q}} = \mathbf{B}(\mathbf{q}) \boldsymbol{\Gamma} + \boldsymbol{\Gamma}_d,$$

where $\mathbf{B}(\mathbf{q}) = [\mathbf{b}_1(\mathbf{q}), \dots, \mathbf{b}_m(\mathbf{q})] \in \mathbb{R}^{n \times m}$ is the input coupling matrix (also called external force matrix), $\boldsymbol{\Gamma} \in \mathbb{U}$ is the control action vector, being $\mathbb{U}$ the m -dimensional actuation space, and $\boldsymbol{\Gamma}_d$ represents the total effect of the modeling errors and energy-bounded external disturbances acting on the system.

Considering this expression, the equation of motion can be expressed by the following form (Kelly et al., 2005):

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \dot{\mathbf{M}}(\mathbf{q}) \dot{\mathbf{q}} - \frac{1}{2} \frac{\partial}{\partial \mathbf{q}} (\dot{\mathbf{q}}' \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}) + \frac{\partial \mathcal{U}(\mathbf{q})}{\partial \mathbf{q}} = \mathbf{B}(\mathbf{q}) \boldsymbol{\Gamma} + \boldsymbol{\Gamma}_d,$$

or, in a compact matrix form, as follows:

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B}(\mathbf{q}) \boldsymbol{\Gamma} + \boldsymbol{\Gamma}_d, \quad (2.36)$$

where $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}}$ is the so-called Coriolis and centrifugal force vector and $\mathbf{G}(\mathbf{q})$ represents the gravitational force vector.

The Coriolis and centrifugal force matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is not unique, but the vector $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ is indeed unique. For the sake of convenience, in the thesis this matrix is obtained through the well-known Christoffel Symbols (of the first kind), $c_{ijk}(\mathbf{q})$, defined as follows:

$$c_{ijk} := \frac{1}{2} \left\{ \frac{\partial m_{kj}}{\partial q_i} + \frac{\partial m_{ki}}{\partial q_j} - \frac{\partial m_{ij}}{\partial q_k} \right\}, \quad (2.37)$$

and the (k, j) th element of the matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is given by:

$$\begin{aligned} c_{kj} &= \sum_{i=1}^n c_{ijk}(\mathbf{q}) \dot{q}_i \\ &= \sum_{i=1}^n \frac{1}{2} \left\{ \frac{\partial m_{kj}}{\partial q_i} + \frac{\partial m_{ki}}{\partial q_j} - \frac{\partial m_{ij}}{\partial q_k} \right\} \dot{q}_i. \end{aligned} \quad (2.38)$$

Furthermore, the equation (2.36) can be represented by the following recursive expression:

$$\sum_{j=1}^n m_{kj}(\mathbf{q}) \ddot{q}_j + \sum_{i=1}^n \sum_{j=1}^n c_{ijk}(\mathbf{q}) \dot{q}_i \dot{q}_j + g_k(\mathbf{q}) = e'_k \mathbf{B}(\mathbf{q}) \boldsymbol{\Gamma} + \boldsymbol{\Gamma}_d, \quad k = 1, \dots, n, \quad (2.39)$$

where e_k is the k th standard basis in $\mathfrak{R}^n$, $g_k(\mathbf{q}) = \frac{\partial \mathcal{U}(\mathbf{q})}{\partial q_k}$.

The equations (2.39) have three types of terms. The first one involves the second derivative of the generalized coordinates. The second part involves quadratic terms in the first derivative of $\mathbf{q}$, where the coefficients may depend on $\mathbf{q}$. These latter terms are further classified into those involving a product of the type $\dot{q}_i^2$ and those involving a product of the type $\dot{q}_i \dot{q}_j$ where $i \neq j$. Terms of the type $\dot{q}_i^2$ are called centrifugal, while terms of the type $\dot{q}_i \dot{q}_j$ are called Coriolis terms. The third type of terms are those involving only $\mathbf{q}$ but not its derivative. This third type arises from differentiating the potential energy.

It is interesting to mention here some properties of the Coriolis and centrifugal force vector $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$, which helps at the control design stage and to perform the stability analysis, i.e.:

1. Property 1: $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ is quadratic in $\dot{\mathbf{q}}$.
2. Property 2: $\|\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}\| \leq v_b \|\dot{\mathbf{q}}\|^2$.
3. Property 3: The matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, obtained as in (2.38), is related to the inertia

matrix $\dot{\mathbf{M}}(\mathbf{q})$ by the expression (Kelly et al., 2005):

$$\mathbf{x}' \left(\frac{1}{2} \dot{\mathbf{M}}(\mathbf{q}) - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \right) \mathbf{x} = 0 \quad \forall \mathbf{q}, \dot{\mathbf{q}}, \mathbf{x} \in \mathbb{Q},$$

and consequently $\mathcal{N}(\mathbf{q}, \dot{\mathbf{q}}) = \dot{\mathbf{M}}(\mathbf{q}) - 2\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is a skew-symmetric matrix, and the following expression holds:

$$\dot{\mathbf{M}}(\mathbf{q}) = \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})'.$$

This property states that the fictitious forces $\mathcal{N}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ do no work in the mechanical system (Lewis et al., 2004).

Moreover, the gravity vector is also bounded, so that $\|\mathbf{G}(\mathbf{q})\| \leq g_b$.

2.4.1 Model Uncertainties

Since the equations of motion (2.36) represent the real system, the dynamic model can be divided in nominal and uncertain parts. Thus, the system can be rewritten as follows:

$$\hat{\mathbf{M}}(\mathbf{q})\ddot{\mathbf{q}} + \hat{\mathbf{C}}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \hat{\mathbf{G}}(\mathbf{q}) = \hat{\mathbf{B}}(\mathbf{q})\boldsymbol{\Gamma} + \boldsymbol{\delta}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\Gamma}_d), \quad (2.40)$$

where $\hat{\mathbf{M}}(\mathbf{q})$, $\hat{\mathbf{C}}(\mathbf{q}, \dot{\mathbf{q}})$ and $\hat{\mathbf{G}}(\mathbf{q})$ are the nominal matrices and vectors. The term $\boldsymbol{\delta}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\Gamma}_d)$ considers the uncertainties of the model associated to an imperfect knowledge of the physical parameters that characterize the system, modeling errors, unmodeled dynamics of the actuators, sensors or structural mechanical vibrations, friction phenomena, electrical noise signals, computational errors and exogenous disturbances (Vivas, 2004). Besides, this term can be represented by additive uncertainties and partitioned into two terms as follows:

$$\boldsymbol{\delta}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\Gamma}_d) = \boldsymbol{\Gamma}_d + \boldsymbol{\Xi}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}), \quad (2.41)$$

where $\boldsymbol{\Gamma}_d$ represents the unmodeled dynamics and energy-bounded external disturbances, as assumed before, while $\boldsymbol{\Xi}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ is a function of the parametric uncertainties of the system matrices and vectors.

From the equations of motion (2.36) and (2.40), the uncertain part of the dy-

dynamic model can be expressed in the following form:

$$\begin{aligned}\Delta\mathbf{M}(\mathbf{q}) &:= \widehat{\mathbf{M}}(\mathbf{q}) - \mathbf{M}(\mathbf{q}), \\ \Delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) &:= \widehat{\mathbf{C}}(\mathbf{q}, \dot{\mathbf{q}}) - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}), \\ \Delta\mathbf{G}(\mathbf{q}) &:= \widehat{\mathbf{G}}(\mathbf{q}) - \mathbf{G}(\mathbf{q}),\end{aligned}$$

being $\Delta\mathbf{M}(\mathbf{q})$, $\Delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ and $\Delta\mathbf{G}(\mathbf{q})$ the parametric uncertainties. Thus, function $\mathbf{\Xi}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ is given by:

$$\mathbf{\Xi}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) = \Delta\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \Delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \Delta\mathbf{G}(\mathbf{q}). \quad (2.42)$$

In what follows, the equations of motion of the *QuadRotor* helicopter will be developed using the Euler-Lagrange formulation presented in this section.

2.5 Helicopter Model via Euler-Lagrange Formulation

The helicopter motion equations can be expressed by the Euler-Lagrange formalism based on the kinetic and potential energy concept:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})}{\partial \mathbf{q}} = \begin{bmatrix} \mathbf{f}_\xi \\ \boldsymbol{\tau}_\eta \end{bmatrix}, \quad (2.43)$$

where $\boldsymbol{\tau}_\eta = \boldsymbol{\tau}_{\eta_a} + \boldsymbol{\delta}_\eta(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau}_{\eta_d}) \in \mathbb{R}^3$ represents the total rolling, pitching and yawing moments expressed in the inertial reference frame, which join the applied torque vector $\boldsymbol{\tau}_{\eta_a}$ and the torques generated by the total effect of the modeling errors of the system and external disturbances. Recalling the equation (2.41), the uncertainty vector for the rotational motion is defined by:

$$\boldsymbol{\delta}_\eta(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau}_{\eta_d}) = \boldsymbol{\tau}_{\eta_d} + \mathbf{\Xi}_\eta(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}), \quad (2.44)$$

where $\boldsymbol{\tau}_{\eta_d} = \mathbf{a}_R + \boldsymbol{\tau}_{G_{\mathcal{J}}}$ represents the external disturbances, which, in this thesis, are assumed to be composed by the gyroscopic movements and the aerodynamic moment vector that affect the helicopter, both expressed in $\mathcal{J}$. The aerodynamic moment vector acting on the helicopter, $\mathbf{a}_R = [a_\phi \ a_\theta \ a_\psi]'$, could be computed depending of the aerodynamic coefficients, air density and squared velocity of the helicopter with respect to the air. However, these moments are unknown in the presence of unpredictable winds and turbulence. Besides, the gyroscopic effects

are considered also unknown, because it is assumed, at the motion control design stage, that there are no access to the rotor speed. Consequently, these effects will be neglected for the control design and will be considered as external disturbances.

The translational force vector $\mathbf{f}_\xi = \mathbf{R}_{\mathcal{I}} \hat{\mathbf{f}} + \boldsymbol{\delta}_\xi(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{f}_{\xi_d}) \in \mathbb{R}^3$ is also divided into two parts: the first term, $\mathbf{R}_{\mathcal{I}} \hat{\mathbf{f}} = \mathbf{R}_{\mathcal{I}e_3} T$, constitutes the applied force to the helicopter due to the main control input T in the $\vec{z}$ axis direction². The second part, $\boldsymbol{\delta}_\xi(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{f}_{\xi_d})$, combines the parametric uncertainties with the external disturbances. For control design purposes, aerodynamic force vector, $\mathbf{f}_{\xi_d} = \mathbf{a}_T = [a_x \ a_y \ a_z]'$, whose components are in the $\vec{x}$, $\vec{y}$ and $\vec{z}$ axes respectively, are also assumed external disturbances. This force could be computed in the same way as in the case of aerodynamic moments.

The generalized coordinates of a rigid body evolving in a three-dimensional space can be written as follows:

$$\mathbf{q} = [\boldsymbol{\xi}' \ \boldsymbol{\eta}']' \in \mathbb{R}^6,$$

with $\boldsymbol{\xi} = [x \ y \ z]'$ $\in \mathbb{R}^3$ being the center of rotation position of the helicopter with respect to the inertial frame $\mathcal{I}$, and $\boldsymbol{\eta} = [\phi \ \theta \ \psi]'$ $\in \mathbb{R}^3$ represents the Euler angles described in Section 2.2.

To derive the equations of motion, first the kinetic and potential energy must be computed in terms of the generalized coordinates of the system. To start it, the inertial, the rotation body-fixed and the center of mass frames are invoked again (see Fig. 2.2).

First, the kinetic energy is developed considering the point $\mathbf{p}$ rigidly attached to the frame $\mathcal{C}$, with coordinates given by equation (2.10) expressed in the frame $\mathcal{I}$, and linear velocity vector also expressed in the inertial frame obtained by equation (2.29). Thus, let $\mathcal{K}$ be the kinetic energy at point $\mathbf{p}$ with respect to the frame $\mathcal{I}$, and let $d\mathcal{K}$ be the kinetic energy of a particle with differential mass dm in $\mathcal{C}$. The kinetic energy of the rigid body as a quadratic function of the first time derivative of the generalized coordinates is obtained by:

$$d\mathcal{K} = \frac{1}{2} (\mathbf{v}_{\mathcal{I}}' \cdot \mathbf{v}_{\mathcal{I}}) dm, \quad (2.45)$$

²The notation $\mathbf{e}_3$ represents the vector $\mathbf{e}_3 = [0 \ 0 \ 1]'$. Thus, the term $\mathbf{R}_{\mathcal{I}e_3}$ denotes the third column of the rotation matrix.

and the squared velocity is given by:

$$\begin{aligned}
\mathbf{v}_{\mathcal{J}}' \cdot \mathbf{v}_{\mathcal{J}} &= \left(\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p} + \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r} + \dot{\boldsymbol{\xi}} \right)' \cdot \left(\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p} + \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r} + \dot{\boldsymbol{\xi}} \right) \\
&= \dot{\boldsymbol{\xi}}' \dot{\boldsymbol{\xi}} + 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p} + 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r} \\
&\quad + (\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p})' (\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p}) + 2(\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times {}^{\mathcal{C}}\mathbf{p})' (\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r}) \\
&\quad + (\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r})' (\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r}) .
\end{aligned}$$

By making use of the skew-symmetric matrix (2.22) and some of the cross product properties, the squared of the velocity is rearranged as follows:

$$\begin{aligned}
\mathbf{v}_{\mathcal{J}}' \cdot \mathbf{v}_{\mathcal{J}} &= \dot{\boldsymbol{\xi}}' \dot{\boldsymbol{\xi}} - 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) \boldsymbol{\omega} - 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}(\mathbf{r}) \boldsymbol{\omega} \\
&\quad + \boldsymbol{\omega}' (\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) + 2\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}(\mathbf{r}) + \mathbf{S}(\mathbf{r})' \mathbf{S}(\mathbf{r})) \boldsymbol{\omega} .
\end{aligned}$$

However, it is more suitable to write this expression in terms of the generalized coordinates. To perform that, the vector $\boldsymbol{\omega}$ is replaced by equation (2.25), which yields:

$$\begin{aligned}
\mathbf{v}_{\mathcal{J}}' \cdot \mathbf{v}_{\mathcal{J}} &= \dot{\boldsymbol{\xi}}' \dot{\boldsymbol{\xi}} - 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} - 2\dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}(\mathbf{r}) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} \\
&\quad + \dot{\boldsymbol{\eta}}' \mathbf{W}_{\eta}' (\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) + 2\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}(\mathbf{r}) + \mathbf{S}(\mathbf{r})' \mathbf{S}(\mathbf{r})) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} .
\end{aligned} \tag{2.46}$$

Therefore, solving the equation (2.45) with the squared velocity, the kinetic energy of the helicopter can be computed by the following expression:

$$\begin{aligned}
\mathcal{K} &= \frac{1}{2} \int \mathbf{v}_{\mathcal{J}}' \cdot \mathbf{v}_{\mathcal{J}} dm , \\
\mathcal{K} &= \frac{1}{2} \int \dot{\boldsymbol{\xi}}' \dot{\boldsymbol{\xi}} dm - \int \dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} dm - \int \dot{\boldsymbol{\xi}}' \mathbf{R}_{\mathcal{J}} \mathbf{S}(\mathbf{r}) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} dm \\
&\quad + \frac{1}{2} \int \dot{\boldsymbol{\eta}}' \mathbf{W}_{\eta}' (\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}({}^{\mathcal{C}}\mathbf{p}) + 2\mathbf{S}({}^{\mathcal{C}}\mathbf{p})' \mathbf{S}(\mathbf{r}) + \mathbf{S}(\mathbf{r})' \mathbf{S}(\mathbf{r})) \mathbf{W}_{\eta} \dot{\boldsymbol{\eta}} dm .
\end{aligned}$$

By integrating the equation above, the second integral and the second term of

the fourth integral become null, and the kinetic energy results on:

$$\mathcal{K} = \frac{1}{2}m\dot{\boldsymbol{\xi}}'\dot{\boldsymbol{\xi}} - m\dot{\boldsymbol{\xi}}'\mathbf{R}_{\mathcal{J}}\mathbf{S}(\mathbf{r})\mathbf{W}_{\boldsymbol{\eta}}\dot{\boldsymbol{\eta}} + \frac{1}{2}\dot{\boldsymbol{\eta}}'\mathcal{J}(\boldsymbol{\eta})\dot{\boldsymbol{\eta}}, \quad (2.47)$$

or in the matrix form:

$$\mathcal{K} = \frac{1}{2} \begin{bmatrix} \dot{\boldsymbol{\xi}} \\ \dot{\boldsymbol{\eta}} \end{bmatrix}' \underbrace{\begin{bmatrix} m\mathbb{1}_{3 \times 3} & -m\mathbf{R}_{\mathcal{J}}\mathbf{S}(\mathbf{r})\mathbf{W}_{\boldsymbol{\eta}} \\ -m\mathbf{W}_{\boldsymbol{\eta}}'\mathbf{S}(\mathbf{r})'\mathbf{R}_{\mathcal{J}}' & \mathcal{J}(\boldsymbol{\eta}) \end{bmatrix}}_{\mathbf{M}(\mathbf{q})} \begin{bmatrix} \dot{\boldsymbol{\xi}} \\ \dot{\boldsymbol{\eta}} \end{bmatrix}, \quad (2.48)$$

where $\mathbf{M}(\mathbf{q})$ is the inertia matrix, and:

$$\mathcal{J}(\boldsymbol{\eta}) = \mathbf{W}_{\boldsymbol{\eta}}'\mathbf{J}\mathbf{W}_{\boldsymbol{\eta}}. \quad (2.49)$$

Assuming the helicopter as a continuous rigid body with mass density $\rho({}^{\mathcal{C}}\mathbf{p})$ at point $\mathbf{p}$ located at its center of mass and displaced by a position $\mathbf{r}$ from the rotation body-fixed frame, the mass of the helicopter is given by:

$$m = \int_V \rho({}^{\mathcal{C}}\mathbf{p}) dx_C dy_C dz_C, \quad (2.50)$$

where V denotes the volume of the body. By using this equation and the Steiner's parallel axis theorem, the moment of inertia tensor $\mathbf{J}$ is given by:

$$\begin{aligned} \mathbf{J} &= \int (\mathbf{S}({}^{\mathcal{C}}\mathbf{p})'\mathbf{S}({}^{\mathcal{C}}\mathbf{p}) + \mathbf{S}(\mathbf{r})'\mathbf{S}(\mathbf{r})) dm \\ &= \int_V (\mathbf{S}({}^{\mathcal{C}}\mathbf{p})'\mathbf{S}({}^{\mathcal{C}}\mathbf{p}) + \mathbf{S}(\mathbf{r})'\mathbf{S}(\mathbf{r})) dx_C dy_C dz_C \\ &= \mathbf{I} + m\mathbf{S}(\mathbf{r})'\mathbf{S}(\mathbf{r}), \end{aligned} \quad (2.51)$$

where:

$$\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix} = \int_V \begin{bmatrix} z_C^2 + y_C^2 & -x_C y_C & -x_C z_C \\ -x_C y_C & x_C^2 + z_C^2 & -z_C y_C \\ -x_C z_C & -z_C y_C & x_C^2 + y_C^2 \end{bmatrix} dx_C dy_C dz_C, \quad (2.52)$$

and

$$m\mathbf{S}(\mathbf{r})'\mathbf{S}(\mathbf{r}) = m \begin{bmatrix} r_z^2 + r_y^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_z r_y \\ -r_x r_z & -r_z r_y & r_x^2 + r_y^2 \end{bmatrix}. \quad (2.53)$$

To complete the Lagrangian, the potential energy term must be computed. In the case of the helicopter, the source of potential energy is the gravity. The potential energy of the helicopter can be computed by assuming the equation of the rigid motion (2.10) and is given by:

$$\begin{aligned}\mathcal{U} &= \int \mathbf{g}'^{\mathcal{I}} \mathbf{p} dm \\ &= m \mathbf{g}'^{\mathcal{I}} \mathbf{p},\end{aligned}\quad (2.54)$$

where $\mathbf{g}$ is the vector of the direction of gravity expressed in the inertial frame, which in this case $\mathbf{g} = [0 \quad 0 \quad -g]'$. Thereby, the potential energy takes the following form:

$$\mathcal{U} = mg(-r_x \sin \theta + r_y \cos \theta \sin \phi + r_z \cos \theta \cos \phi + z). \quad (2.55)$$

Once, the kinetic and potential energy of the *QuadRotor* helicopter have been computed (see (2.48) and (2.55)), the Lagrangian (2.31) of the system can be written as follows:

$$\begin{aligned}\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) &= \mathcal{K}(\mathbf{q}, \dot{\mathbf{q}}) - \mathcal{U}(\mathbf{q}) \\ &= \frac{1}{2} \dot{\mathbf{q}}' \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} - m \mathbf{g}'^{\mathcal{I}} \mathbf{p}.\end{aligned}\quad (2.56)$$

By solving the derivatives required by the Euler-Lagrange equations (2.35), the equations of motion, expressed in the form of the equation (2.40)³, results in:

$$\begin{aligned}\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) &= \mathbf{B}(\mathbf{q})\boldsymbol{\Gamma} + \boldsymbol{\delta}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\Gamma}_d) \\ \left[\begin{array}{cc} m\mathbf{1}_{3 \times 3} & -m\mathbf{R}_{\mathcal{I}}\mathbf{S}(\mathbf{r})\mathbf{W}_{\eta} \\ -m\mathbf{W}'_{\eta}\mathbf{S}(\mathbf{r})'\mathbf{R}_{\mathcal{I}}' & \mathcal{J}(\boldsymbol{\eta}) \end{array} \right] \cdot \left[\begin{array}{c} \ddot{\boldsymbol{\xi}} \\ \ddot{\boldsymbol{\eta}} \end{array} \right] + \left[\begin{array}{cc} \mathbf{C}_{\xi\xi}(\mathbf{q}, \dot{\mathbf{q}}) & \mathbf{C}_{\xi\eta}(\mathbf{q}, \dot{\mathbf{q}}) \\ \mathbf{C}_{\eta\xi}(\mathbf{q}, \dot{\mathbf{q}}) & \mathbf{C}_{\eta\eta}(\mathbf{q}, \dot{\mathbf{q}}) \end{array} \right] \cdot \left[\begin{array}{c} \dot{\boldsymbol{\xi}} \\ \dot{\boldsymbol{\eta}} \end{array} \right] \\ + \left[\begin{array}{c} \mathbf{G}_{\xi}(\mathbf{q}) \\ \mathbf{G}_{\eta}(\mathbf{q}) \end{array} \right] &= \left[\begin{array}{c} \mathbf{B}_{\xi}(\mathbf{q}) \\ \mathbf{B}_{\eta}(\mathbf{q}) \end{array} \right] \cdot \left[\begin{array}{c} T \\ \boldsymbol{\tau}_{\eta_a} \end{array} \right] + \left[\begin{array}{c} \boldsymbol{\delta}_{\xi}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, f_{\xi_d}) \\ \boldsymbol{\delta}_{\eta}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau}_{\eta_d}) \end{array} \right].\end{aligned}\quad (2.57)$$

where the elements of the Coriolis and centrifugal matrix can be computed by the Christoffel symbols using the equation (2.38). The gravitational vector has the

³For the sake of notation simplicity, the matrices and vectors of the nominal system will be presented without the argument $\hat{\cdot}$. Throughout the thesis, where necessary to make distinction with regard to this simplification, it will be observed.

following form:

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} 0 \\ 0 \\ mg \\ -mg(-r_y \cos \theta \cos \phi + r_z \cos \theta \sin \phi) \\ -mg(r_x \cos \theta + r_y \sin \theta \sin \phi + r_z \sin \theta \cos \phi) \\ 0 \end{bmatrix}, \quad (2.58)$$

while the force matrix is given by:

$$\mathbf{B}(\mathbf{q}) = \begin{bmatrix} \mathbf{B}_\xi(\mathbf{q}) \\ \dots \\ \mathbf{B}_\eta(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{\mathcal{J}_{e_3}} & \mathbf{0}_{3 \times 3} \\ \dots & \dots \\ \mathbf{0}_{3 \times 1} & \mathbf{1}_{3 \times 3} \end{bmatrix}. \quad (2.59)$$

In summary, the equations of motion obtained in this section provide a strongly coupled dynamic model. This model will be used to emulate a real *QuadRotor* helicopter. In the following section this dynamic model is simplified and a decentralized model is achieved.

2.5.1 Simplified Equations of Motion of the Helicopter

In this section, the center of mass and the origin of the rotation body-fixed frame $\mathcal{B}$ are assumed congruent, which results in the vector $\mathbf{r} = 0$.

Consequently, the kinetic energy terms combining $\dot{\xi}$ and $\dot{\eta}$ in the the Lagrangian (2.56) disappear and, thus, the Euler-Lagrange equations can be divided into two interconnected dynamics, the translational and rotational subsystems. These simplified equations of motion compose a decentralized dynamic model, as it can be seen in Fig. 2.3.

Under this assumption, the equations of motion (2.57) can be rewritten as follows:

$$\begin{aligned} \begin{bmatrix} m\mathbf{1}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathcal{J}(\boldsymbol{\eta}) \end{bmatrix} \cdot \begin{bmatrix} \ddot{\xi} \\ \ddot{\eta} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{C}_{\eta\eta}(\mathbf{q}, \dot{\mathbf{q}}) \end{bmatrix} \cdot \begin{bmatrix} \dot{\xi} \\ \dot{\eta} \end{bmatrix} + \begin{bmatrix} mg\mathbf{e}_3 \\ \mathbf{0}_{3 \times 3} \end{bmatrix} \\ = \begin{bmatrix} \mathbf{B}_\xi(\mathbf{q}) \\ \mathbf{B}_\eta(\mathbf{q}) \end{bmatrix} \cdot \begin{bmatrix} T \\ \boldsymbol{\tau}_{\eta_a} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\delta}_\xi(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, f_{\xi_d}) \\ \boldsymbol{\delta}_\eta(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau}_{\eta_d}) \end{bmatrix}, \end{aligned} \quad (2.60)$$

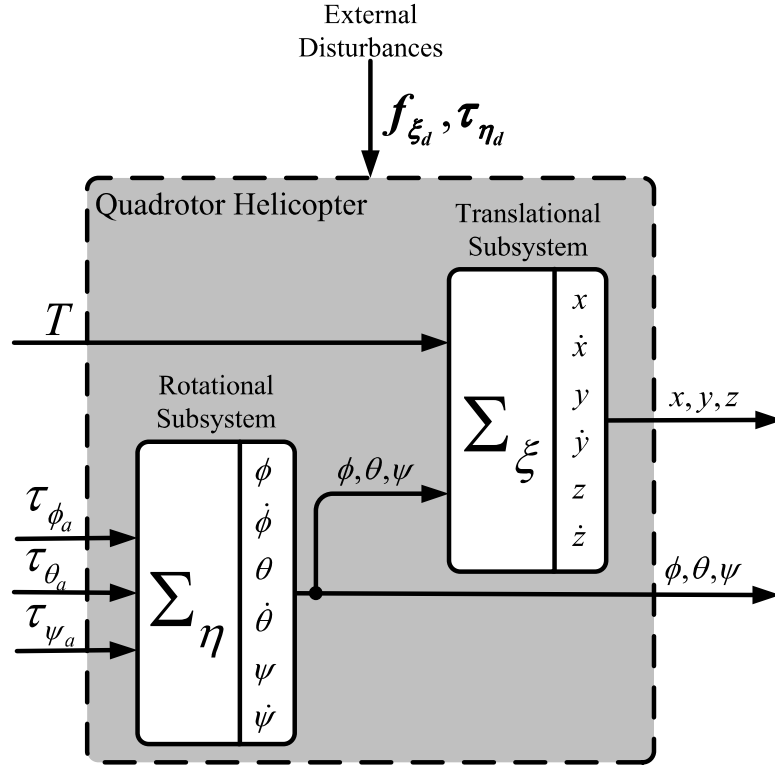


Figure 2.3: Dynamic system divided into two interconnected subsystems.

where, in this model, the inertia matrix of the rotational subsystem is given by $\mathcal{J}(\boldsymbol{\eta}) = \mathbf{W}'_{\boldsymbol{\eta}} \mathbf{I} \mathbf{W}_{\boldsymbol{\eta}}$. From now on, the Coriolis and centrifugal matrix of the rotational subsystem is defined by $\mathbf{C}_{\boldsymbol{\eta}\boldsymbol{\eta}}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}})$.

The *translational movement* can be expressed by means of the state vector $\boldsymbol{\xi}$ by the following equations:

$$\begin{cases} \ddot{x} = \frac{1}{m} (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) T + \frac{\delta_{\xi_x}}{m} \\ \ddot{y} = \frac{1}{m} (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) T + \frac{\delta_{\xi_y}}{m} \\ \ddot{z} = -g + \frac{1}{m} (\cos \theta \cos \phi) T + \frac{\delta_{\xi_z}}{m} \end{cases} \quad (2.61)$$

The simplified rotational equations of motion in terms of $\boldsymbol{\eta}$ can be written as follows:

$$\mathcal{J}(\boldsymbol{\eta}) \ddot{\boldsymbol{\eta}} + \mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) \dot{\boldsymbol{\eta}} = \boldsymbol{\tau}_{\boldsymbol{\eta}}, \quad (2.62)$$

where:

$$\mathcal{J}(\boldsymbol{\eta}) = \begin{bmatrix} I_{xx} & 0 & -I_{xx}S\theta \\ 0 & I_{yy}C^2\phi + I_{zz}S^2\phi & (I_{yy} - I_{zz})C\phi S\phi C\theta \\ -I_{xx}S\theta & (I_{yy} - I_{zz})C\phi S\phi C\theta & I_{xx}S^2\theta + I_{yy}S^2\phi C^2\theta + I_{zz}C^2\phi C^2\theta \end{bmatrix}, \quad (2.63)$$

and

$$\mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix},$$

with:

$$c_{11} = 0$$

$$c_{12} = (I_{yy} - I_{zz})(\dot{\theta}C\phi S\phi + \frac{1}{2}\dot{\psi}S^2\phi C\theta) + \frac{1}{2}(I_{zz} - I_{yy})\dot{\psi}C^2\phi C\theta - \frac{1}{2}I_{xx}\dot{\psi}C\theta$$

$$c_{13} = (I_{zz} - I_{yy})(\dot{\psi}C\phi S\phi C^2\theta + \frac{1}{2}\dot{\theta}C\theta C^2\phi) + \frac{1}{2}(I_{yy} - I_{zz})\dot{\theta}C\theta S^2\phi - \frac{1}{2}I_{xx}\dot{\theta}C\theta$$

$$c_{21} = (I_{zz} - I_{yy})(\dot{\theta}C\phi S\phi + \frac{1}{2}\dot{\psi}S^2\phi C\theta) + \frac{1}{2}(I_{yy} - I_{zz})\dot{\psi}C^2\phi C\theta + \frac{1}{2}I_{xx}\dot{\psi}C\theta$$

$$c_{22} = (I_{zz} - I_{yy})\dot{\phi}C\phi S\phi$$

$$c_{23} = -I_{xx}\dot{\psi}S\theta C\theta + I_{yy}\dot{\psi}S^2\phi C\theta S\theta + I_{zz}\dot{\psi}C^2\phi S\theta C\theta + \frac{1}{2}(I_{yy} - I_{zz})\dot{\phi}C\theta C^2\phi \\ + \frac{1}{2}(I_{zz} - I_{yy})\dot{\phi}C\theta S^2\phi + \frac{1}{2}I_{xx}\dot{\phi}C\theta$$

$$c_{31} = (I_{yy} - I_{zz})(\dot{\psi}C^2\theta S\phi C\phi + \frac{1}{2}\dot{\theta}C\theta C^2\phi) + \frac{1}{2}(I_{zz} - I_{yy})\dot{\theta}C\theta S^2\phi - \frac{1}{2}I_{xx}\dot{\theta}C\theta$$

$$c_{32} = (I_{zz} - I_{yy})(\dot{\theta}C\phi S\phi S\theta + \frac{1}{2}\dot{\phi}S^2\phi C\theta) + \frac{1}{2}(I_{yy} - I_{zz})\dot{\phi}C^2\phi C\theta - \frac{1}{2}I_{xx}\dot{\phi}C\theta \\ + I_{xx}\dot{\psi}S\theta C\theta - I_{yy}\dot{\psi}S^2\phi S\theta C\theta - I_{zz}\dot{\psi}C^2\phi S\theta C\theta$$

$$c_{33} = (I_{yy} - I_{zz})\dot{\phi}C\phi S\phi C^2\theta - I_{yy}\dot{\theta}S^2\phi C\theta S\theta - I_{zz}\dot{\theta}C^2\phi C\theta S\theta \\ + I_{xx}\dot{\theta}C\theta S\theta.$$

Therefore, the simplified mathematical model that describes the helicopter

rotational movement obtained from the Euler-Lagrange formalism is given by:

$$\ddot{\boldsymbol{\eta}} = \mathcal{J}(\boldsymbol{\eta})^{-1} (\boldsymbol{\tau}_{\boldsymbol{\eta}} - \mathbf{C}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) \dot{\boldsymbol{\eta}}) . \quad (2.64)$$

These two interconnected subsystems will be used to design all control strategies presented in the thesis.

2.6 Helicopter Model via Newton-Euler Formulation

In this section the equations of motion of the *QuadRotor* helicopter are obtained via Newton-Euler formulation. This model is presented just as an alternative approach. Besides, this approach provides a clear understanding of the forces and torques applied to the vehicle, and from this a relationship between these forces and torques and the generalized forces and torques obtained from the Euler-Lagrange dynamic model are obtained. The basis theory of this formulation is omitted. For more details see the references therein the thesis.

The dynamic equations of a rigid body subject to external forces applied to the center of mass and expressed in the body-fixed frame can be obtained through the Newton-Euler approach using the relations (2.22) and (2.30) as follows:

$$\begin{bmatrix} m\mathbb{1}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{J} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}}_{\mathcal{B}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\omega} \times m\mathbf{v}_{\mathcal{B}} \\ \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{\mathcal{B}} \\ \boldsymbol{\tau}_{\mathcal{B}} \end{bmatrix}, \quad (2.65)$$

where $\mathbf{J} \in \mathbb{R}^{3 \times 3}$ is the moment of inertia tensor obtained by equation (2.51), $\mathbb{1}_{3 \times 3} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{0}_{3 \times 3} \in \mathbb{R}^{3 \times 3}$ are the identity and zeros matrices, respectively. $\mathbf{v}_{\mathcal{B}}$ is the linear velocity vector and $\boldsymbol{\omega}$ is the angular rate, both expressed in $\mathcal{B}$.

Considering the state vector $[\boldsymbol{\xi} \quad \mathbf{v}_{\mathcal{J}} \quad \boldsymbol{\eta} \quad \boldsymbol{\omega}]'$, the helicopter dynamic model, assuming the center of mass at the origin of the frame $\mathcal{C}$, can be rewritten by:

$$\begin{cases} \dot{\boldsymbol{\xi}} = \mathbf{v}_{\mathcal{J}} - \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r} \\ m\dot{\mathbf{v}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \mathbf{f}_{\mathcal{B}} \\ \dot{\mathbf{R}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \mathbf{S}(\boldsymbol{\omega}) \\ \mathbf{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} + \boldsymbol{\tau}_{\mathcal{B}} \end{cases} \quad (2.66)$$

where the time derivative of the linear velocity $\mathbf{v}_{\mathcal{J}}$ (see equation (2.29)) with the center of mass at the origin of frame $\mathcal{C}$ (i.e. ${}^{\mathcal{C}}\mathbf{p} = 0$) is given by:

$$\dot{\mathbf{v}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + \mathbf{R}_{\mathcal{J}} \dot{\boldsymbol{\omega}} \times \mathbf{r} + \ddot{\boldsymbol{\xi}}. \quad (2.67)$$

The forces and torques in which the helicopter is subject, $\mathbf{f}_{\mathcal{B}} \in \mathcal{B}$ and $\boldsymbol{\tau}_{\mathcal{B}} \in \mathcal{B}$, consist of: its weight, the aerodynamic force and moment vectors, the gyroscopic effects caused by the rotors, the applied thrust, and the torques generated by the four propellers. These forces and torques can be expressed by the following form:

$$\begin{cases} \mathbf{R}_{\mathcal{J}} \mathbf{f}_{\mathcal{B}} = -mg\mathbf{e}_3 + \mathbf{R}_{\mathcal{J}_{e_3}} T + \mathbf{f}_{\xi_d} \\ \boldsymbol{\tau}_{\mathcal{B}} = \boldsymbol{\tau}_a + \mathbf{a}_r + \boldsymbol{\tau}_G - \mathbf{r} \times (\hat{\mathbf{f}} + \mathbf{R}_{\mathcal{J}'} \mathbf{f}_{\xi_d}) \end{cases} \quad (2.68)$$

where $\boldsymbol{\tau}_a$ is the applied torque vector defined by equation (2.2), $\boldsymbol{\tau}_G$ is the gyroscopic effect vector given by (2.6), and $\mathbf{a}_r = [a_p \ a_q \ a_r]'$ is the aerodynamic moment vector expressed in the body-fixed frame. Given that the aerodynamic moments and the gyroscopic effects are assumed to be external disturbances, they are considered as the vector $\boldsymbol{\tau}_d = \mathbf{a}_r + \boldsymbol{\tau}_G$. In this model, the parametric uncertainties are omitted.

By substituting the equations of forces and torques (2.68) into the dynamic model (2.66), these equations can be rewritten as follows:

$$\begin{cases} \dot{\boldsymbol{\xi}} = \mathbf{v}_{\mathcal{J}} - \mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times \mathbf{r} \\ m\dot{\mathbf{v}}_{\mathcal{J}} = -mg\mathbf{e}_3 + \mathbf{R}_{\mathcal{J}_{e_3}} T + \mathbf{f}_{\xi_d} \\ \dot{\mathbf{R}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \mathbf{S}(\boldsymbol{\omega}) \\ \mathbf{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} + \boldsymbol{\tau}_a + \boldsymbol{\tau}_d - \mathbf{r} \times (\hat{\mathbf{f}} + \mathbf{R}_{\mathcal{J}'} \mathbf{f}_{\xi_d}) \end{cases} \quad (2.69)$$

By replacing the linear acceleration (2.67) into the helicopter model (2.69), and after some algebraic manipulations, the following Newton-Euler equations of motion are obtained:

$$\begin{cases} \mathbf{R}_{\mathcal{J}_{e_3}} T + \mathbf{f}_{\xi_d} &= m\ddot{\boldsymbol{\xi}} - m\mathbf{R}_{\mathcal{J}} \mathbf{r} \times \dot{\boldsymbol{\omega}} - m\mathbf{R}_{\mathcal{J}} \boldsymbol{\omega} \times (\mathbf{r} \times \boldsymbol{\omega}) + mg\mathbf{e}_3 \\ \boldsymbol{\tau}_a + \boldsymbol{\tau}_d &= \mathbf{J}\dot{\boldsymbol{\omega}} - m\mathbf{r} \times (\mathbf{r} \times \dot{\boldsymbol{\omega}}) + m\mathbf{r} \times \mathbf{R}_{\mathcal{J}'} \ddot{\boldsymbol{\xi}} + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} \\ &\quad - m\mathbf{r} \times (\boldsymbol{\omega} \times (\mathbf{r} \times \boldsymbol{\omega})) + m\mathbf{g}\mathbf{r} \times \mathbf{R}_{\mathcal{J}'} \mathbf{e}_3 \end{cases} \quad (2.70)$$

To obtain the Newton-Euler model (2.70) using the Euler angle parameterization and, consequently, expressed by the generalized coordinates $\mathbf{q} = [\boldsymbol{\xi}' \ \boldsymbol{\eta}']'$, the equation (2.25) and its time derivative must be used together with the relation $\boldsymbol{\omega} \times = \mathbf{S}(\boldsymbol{\omega}) = \mathbf{R}_{\mathcal{J}'} \dot{\mathbf{R}}_{\mathcal{J}}$. These equations are obtained in an elementary way and by direct substitution. Thus, in this thesis it will not be developed. At the same time, it is interesting to mention here the relationship between the torques expressed in the body-fixed frame and in the inertial reference frame, that is:

$$\boldsymbol{\tau}_{\eta} = \mathbf{W}_{\eta}' \boldsymbol{\tau}_{\mathcal{B}}, \quad (2.71)$$

which leads to:

$$\begin{aligned}\boldsymbol{\tau}_{\eta_a} &= \mathbf{W}_{\eta'}' \boldsymbol{\tau}_a, \\ \boldsymbol{\tau}_{\eta_d} &= \mathbf{W}_{\eta'}' \boldsymbol{\tau}_d,\end{aligned}$$

As in the case of the helicopter model via Euler-Lagrange formulation, in this section the simplified model is also obtained. Considering the frames $\mathcal{C}$ and $\mathcal{B}$ congruent, the vector $\mathbf{r}$ is null, and, therefore, the equations (2.69) are rewritten as follows:

$$\begin{cases} \dot{\boldsymbol{\xi}} = \mathbf{v}_{\mathcal{J}} \\ m\dot{\mathbf{v}}_{\mathcal{J}} = -mg\mathbf{e}_3 + \mathbf{R}_{\mathcal{J}\mathbf{e}_3} T + \mathbf{f}_{\boldsymbol{\xi}_d} \\ \dot{\mathbf{R}}_{\mathcal{J}} = \mathbf{R}_{\mathcal{J}} \mathbf{S}(\boldsymbol{\omega}) \\ \mathbf{I}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} + \boldsymbol{\tau}_a + \boldsymbol{\tau}_d \end{cases} \quad (2.72)$$

By defining the following state vector:

$$\boldsymbol{\zeta} = [x \quad y \quad z \quad u_0 \quad v_0 \quad w_0 \quad \phi \quad \theta \quad \psi \quad p \quad q \quad r]^T, \quad (2.73)$$

the simplified model of the helicopter (2.72) can be presented in a state space form:

$$\dot{\boldsymbol{\zeta}} = \begin{cases} \dot{x} = u_0 \\ \dot{y} = v_0 \\ \dot{z} = w_0 \\ \dot{u}_0 = \frac{1}{m} (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \cdot T + \frac{a_x}{m} \\ \dot{v}_0 = \frac{1}{m} (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \cdot T + \frac{a_y}{m} \\ \dot{w}_0 = -g + \frac{1}{m} (\cos \theta \cos \phi) \cdot T + \frac{a_z}{m} \\ \dot{\phi} = p + q \sin \phi \tan \theta + r \cos \phi \tan \theta \\ \dot{\theta} = q \cos \phi - r \sin \phi \\ \dot{\psi} = q \sin \phi \sec \theta + r \cos \phi \sec \theta \\ \dot{p} = \frac{(I_{yy} - I_{zz})}{I_{xx}} qr + \frac{\tau_{a_p}}{I_{xx}} + \frac{\tau_{d_p}}{I_{xx}} \\ \dot{q} = \frac{(I_{zz} - I_{xx})}{I_{yy}} pr + \frac{\tau_{a_q}}{I_{yy}} + \frac{\tau_{d_q}}{I_{yy}} \\ \dot{r} = \frac{(I_{xx} - I_{yy})}{I_{zz}} pq + \frac{\tau_{a_r}}{I_{zz}} + \frac{\tau_{d_r}}{I_{zz}} \end{cases} \quad (2.74)$$

The equations of motion of the *QuadRotor* helicopter obtained via Newton-Euler procedure are not used to design the controllers in this thesis. However, many researchers make use of this approach, and some of their controllers used for results comparison have been designed by means of this model.

2.7 System Design Parameters

This section presents the parameters of a *QuadRotor* helicopter that is being designed in the Automation, Control and Robotic Group, Department of Systems Engineering and Automation at the University of Seville.

The design of the *QuadRotor* helicopter structure is based on the illustration presented in Fig. 2.1. The vehicle will be constituted of four brushless motors AXI 2217/16 Gold Line, which are able to provide a maximum thrust of 12.2 N per each motor-propeller group, assuming blades with dimension of $10'' \times 5''$. An Inertial Measurement Unit (IMU) MTi-G from Xsens will be used to measure the linear accelerations, Euler angles and angular rates. An infrared sensor is considered to estimate the altitude, while to obtain the translational displacement, an on-board visual sensor is being implemented. These data will be treated in an embedded computer based on the PC-104 platform, which will be in charge of guiding the vehicle autonomously.

By previous analysis, some values of the model parameters have already been estimated, whose will be used to carry out all the simulation results in this thesis, and are shown in Table 2.1.

2.8 Conclusions

In this chapter the modeling of the *QuadRotor* helicopter has been presented. The models developed here have been obtained taking into account the control design purposes.

A more realistic modeling has been performed considering the center of mass displaced of the helicopter center of rotation, which generates coupling between the translational and rotational movements. From this coupled model, a simplified and decentralized model has been obtained, which will be used to design the control strategies presented in this work.

The equations of motion have been computed through two approaches, the Euler-Lagrange and the Newton-Euler formulations, allowing a wide range of control structures to be chosen.

Table 2.1: *QuadRotor* helicopter model parameters.

Parameter Description	Parameter	Value
Mass of the <i>QuadRotor</i> helicopter	m	2.24 kg
Distance between the mass center and the rotors	l	0.332 m
Thrust coefficient of the rotors	b	$9.5e - 6 \text{ N s}^2$
Drag coefficient of the rotors	k_τ	$1.7e - 7 \text{ N m s}^2$
Gravitational acceleration	g	9.81 m/s^2
Moment of inertia around the x -axis	I_{xx}	0.0363 Kg.m^2
Moment of inertia around the y -axis	I_{yy}	0.0363 Kg.m^2
Moment of inertia around the z -axis	I_{zz}	0.0615 Kg.m^2
Position of the center of mass in x from the body-fixed frame	r_x	-0.00069 m
Position of the center of mass in y from the body-fixed frame	r_y	-0.0014 m
Position of the center of mass in z from the body-fixed frame	r_z	-0.0311 m